

**Table 7-3. Output Voltage (SEL Pin = Low) Settings
(continued)**

OUTPUT VOLTAGE - $V_{O(1)}$ (SEL = LOW)	R_{CFG3}
2.8 V	10.5 k Ω
3.0 V	13.3 k Ω
3.3 V	16.2 k Ω
3.6 V	20.5 k Ω
4.0 V	24.9 k Ω
4.5 V	30.1 k Ω
5.0 V	36.5 k Ω

7.3.7 SEL Pin

The SEL pin selects which configuration bits control the output voltage.

- When SEL = high, the output voltage $V_{O(2)}$ is set.
- When SEL = low, the output voltage $V_{O(1)}$ is set.

7.3.8 Short-Circuit Protection

7.3.8.1 Current Limit Setting = 'Unlimited'

The device has a inbuilt short circuit protection function to limit the current through Q1. The maximum current that flows is limited by the peak current limit. The output voltage decreases if the load is higher than the peak current limit. If the output voltage falls below 1.25 typically, the short circuit protection is activated. With short circuit protection activated the input current is limited to 26 mA on average.

The device automatically restarts to normal operation after the short condition is removed.

7.3.8.2 Current Limit Setting = 1 mA to 100 mA

The input current limiting function automatically limits current during a short-circuit condition. The device regulates the average input current for as long as the short-circuit condition exists. If the output voltage falls below 1.25 V typically, the short circuit protection is activated. For input current limit settings of 100 mA, 50 mA and 25 mA, the short circuit protection limits the input current to 26 mA on average. For input current limit setting of 10 mA, 5 mA, 2.5 mA, and 1 mA, the short circuit protection limits the input current to slightly above the typical values for each setting. [Table 7-4](#) shows the typical short circuit currents for each input current limit setting.

The device automatically restarts to previous operation after the short condition is removed.

Table 7-4. Typical Input Current During Short Circuit Condition ($V_O < 1.25$ V Typically) for All Input Current Limit Settings

INPUT CURRENT LIMIT SETTING	TYPICAL SHORT CIRCUIT INPUT CURRENT
1 mA	1.2 mA
2.5 mA	2.8 mA
5 mA	5.2 mA
10 mA	12 mA
25 mA	26 mA
50 mA	26 mA
100 mA	26 mA
Unlimited	26 mA

7.3.9 Thermal Shutdown

The device has a thermal shutdown function that disables the device if it gets too hot for correct operation. When the device cools down, it automatically restarts operation after a typical delay of $t_{d(RESTART)} = 10$ ms. The device starts with the soft-start feature (see [Section 7.3.3](#)) and keeps the previously read CFG pin setting.

7.4 Device Functional Modes

The device has two functional modes: on and off. The device enters the on mode when the voltage on the VIN pin is higher than the UVLO threshold and a high logic level is applied to the EN pin. The device enters the off mode when the voltage on the VIN pin is lower than the UVLO threshold or a low logic level is applied to the EN pin.

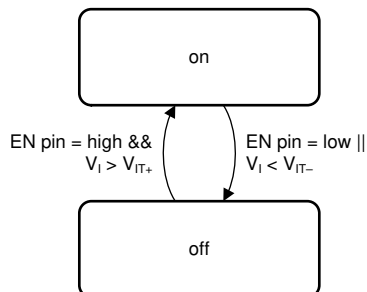


Figure 7-13. Device Functional Modes